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Porichoy (পরিচয়)

DPP-in-a-Box for Bangladesh's garment factories

Project report — Needle Innovation Challenge 3.0 · "Better Process" focus area.

Every garment gets a verifiable identity before the EU's 2027 wall.



At a glance — Live product: rudra496.github.io/porichoy (repo [rudra496/porichoy](https://github.com/rudra496/porichoy)) · 19/19 automated engine tests, CI green · 16 Crossref-verified peer-reviewed references · ≈4 million RMG workers (BGMEA), majority women · EU ESPR in force 18 Jul 2024, textile DPP obligations expected ~2027–30 with customs auto-checks · Applicant: Rudra Sarker, final-year IPE, SUST.

Prepared by Rudra Junior for Rudra Sarker · 2026-09-13 · All claims verified; unverifiable items flagged. Photo: S. Biswas, CC BY-SA 4.0, Wikimedia Commons.

1 · Executive summary

Porichoy ("identity") turns a Bangladeshi garment factory's existing messy records — Excel sheets with mixed Bangla/English headers, ERP exports, paper trim cards — into buyer-ready EU Digital Product Passports, scored by readiness and published as scannable QR passports, powered by women **DPP Data Stewards** trained at each factory.

90/100

readiness achieved by
sample POs on first
ingest (live demo)

19/19

engine unit tests
passing, CI-verified

16

Crossref-verified peer-
reviewed references
(\$8)

2027

expected textile
delegated act;
customs auto-checks
follow

Why now: the EU's Ecodesign for Sustainable Products Regulation (in force 18 July 2024; textiles in the first Working Plan, April 2025) makes Digital Product Passports mandatory for EU market access, with customs performing automatic checks on passport existence and authenticity [1,2]. For the world's #2 RMG exporter whose biggest market is Europe, this is a dated compliance wall. The data factories need already exists — but scattered and unstructured. Global traceability platforms are brand-side and enterprise-priced; nobody captures at the Bangladeshi factory floor. That last mile is Porichoy.

Honesty statement. Stage: Launched/Product-market fit — product live since September 2026, pilot factories in recruitment, pre-revenue by design until the grant-funded 10-factory pilot begins. We claim no revenue, no company registration (planned in the grant period), and no figures we could not verify. Roadmap numbers are labelled targets.

2 · Problem & regulation

The regulatory forcing function is dated and external [1,2]. From 2027–30 DPP obligations phase in for textiles placed on the EU market; EU customs will run automatic checks on passport existence and authenticity. A Bangladeshi factory that cannot produce passport data risks losing EU orders — and with them, the livelihoods of a workforce of ≈4 million (BGMEA), the majority women.

The dated wall: EU Digital Product Passport timeline (European Commission)



Figure 1 — EU DPP timeline. Source: European Commission ESPR page (fetched 2026-09-12); CIRPASS-2 pilots verified on cirpass2.eu (2026-09-13).

Meanwhile the evidence base shows both the traceability gap in textile supply chains [4] and Bangladesh-specific compliance and circularity challenges [5,6,7]: social-compliance performance is a documented constraint of the RMG industry, informal circularity is structurally important, and textile-waste mismanagement carries measurable environmental cost.

3 • The solution

Four steps, all live in the production app: **1 • Ingest** — drop a factory Excel/CSV; every sheet is parsed on-device (local-first: raw records never leave the factory). **2 • Map** — a deterministic mapping engine aligns chaotic headers to the canonical DPP schema using documented synonym patterns (English + Bangla, Bangla numerals transliterated) with a bigram-dice fuzzy fallback; a trained steward confirms; each factory's vocabulary is learned. **3 • Score** — each production order receives a DPP Readiness Score (0–100; weights = ESPR Art. 8 content categories; a valid unique identifier is mandatory). **4 • Passport** — one click publishes a QR whose link embeds a verifiable passport snapshot; scanning opens the buyer view on any phone — no backend, no data leaving the factory.

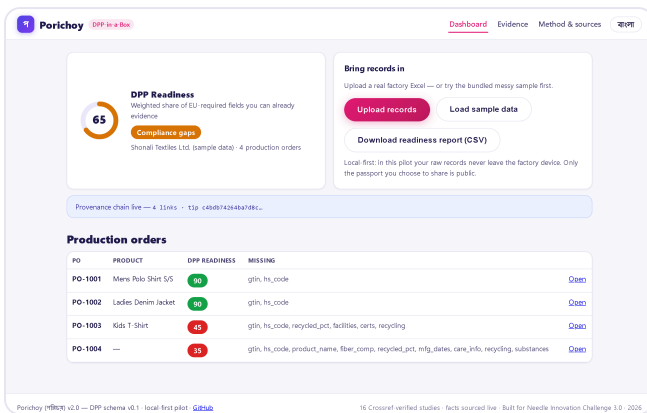


Figure 2 — Live dashboard: per-order readiness from messy sample files (real app).

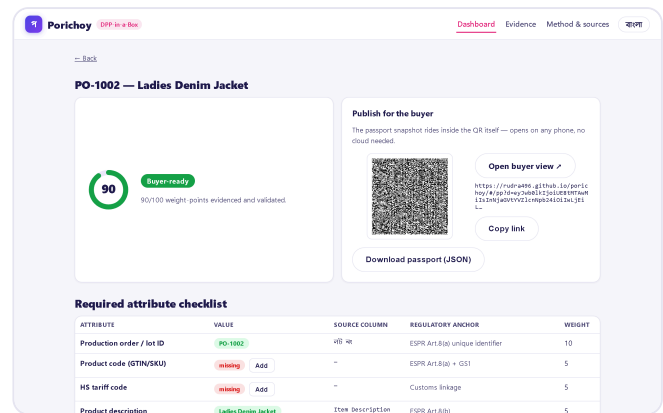


Figure 3 — PO view: attribute checklist with ESPR anchors, source columns, QR publish.

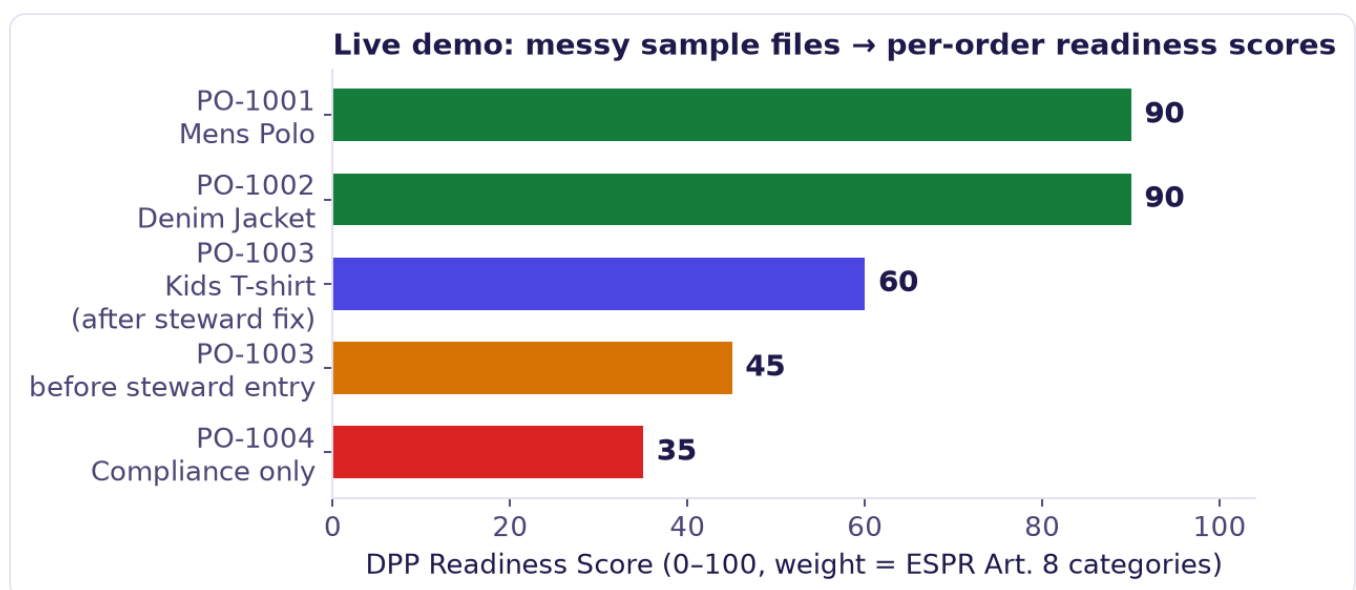


Figure 4 — Readiness outcomes on the bundled messy sample (live demo data). Steward in-place gap-filling lifted PO-1003 from 45 to 60 in the live test; every accepted value extends the SHA-256 provenance chain.

4 • Engineering & verification

Canonical schema v0.1 (versioned)

Twelve attributes anchored to ESPR Article 8 content categories and public CIRPASS-2 textile-pilot taxonomies (verified on cirpass2.eu 2026-09-13), each carrying a regulatory anchor and a Readiness weight summing to 100. The textile delegated act is still a draft — the schema sits behind a versioned mapping layer, so final EU rules are a configuration change, not a rebuild.

Provenance & honesty engineering

Accepted values join an append-only SHA-256 hash chain (hash = SHA256(prevHash | seq | pold | attrsJson | at)); chain verification re-derives every link and detects tampering (unit-tested). Duplicate PO rows merge; conflicting values are flagged to the steward, never silently overwritten. Confidence scores accompany every mapping suggestion. Published QRs are invalidated automatically when underlying data changes, forcing re-publication. 19/19 automated tests cover mapping (Bangla headers, concatenated headers, no-double-claim), validation (percent forms incl. Bangla numerals), merge/conflict semantics, scoring invariants and chain tamper-detection; CI runs on every push.

5 • Market & business model

SEGMENT	PLAYERS	GAP PORICHOY EXPLOITS
Brand-side DPP SaaS	TrusTrace, TextileGenesis, Reverse Resources	Consume clean data; no BD factory-floor capture; enterprise pricing
Consultancies	Big-4 / compliance firms	Per-visit cost; no product; nothing learned or reused
Porichoy's square	The BD last-mile data engine	Sits at the factory, speaks Bangla, feeds any platform — partner, not rival

Revenue: factory SaaS BDT 2,000–8,000/month by size; one-time onboarding & data-rescue fees; brand-funded supplier onboarding (buyers pay to make their supplier base compliant before 2027–30). Onboarding ≈ 2 person-days per factory; software gross margin >80%; replication is a per-factory template. Beachhead: mid-size exporters already receiving buyer DPP requests.

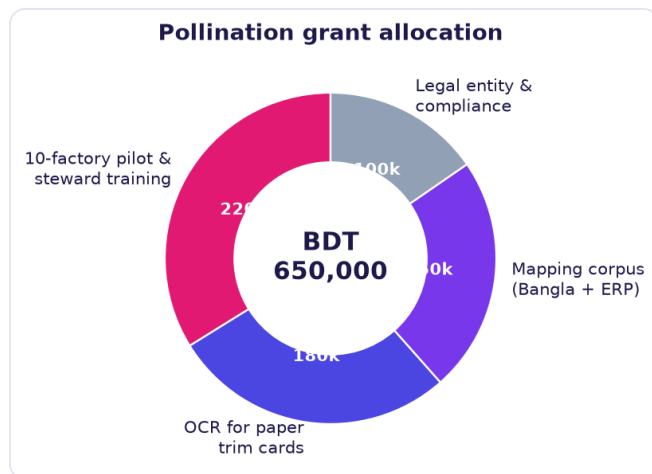


Figure 5 — Pollination grant allocation (sums to BDT 650,000).

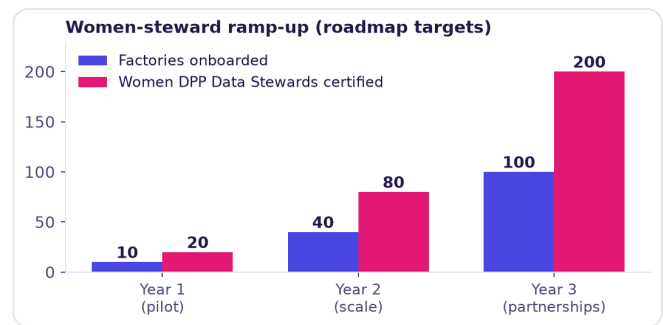


Figure 6 — Women-steward ramp (roadmap targets).

6 · Women at the center

NIC eligibility requires solutions that "ensure future work prospects of women garment workers" — Porichoy is designed around it, not retrofitted: (1) **new skilled jobs** — two women operators per factory trained and certified as DPP Data Stewards, formal higher-skilled back-office roles with a public micro-credential (20 certified in year 1; 200 by year 3 — targets, Fig. 6); (2) **jobs protected at scale** — keeping factories EU-order-eligible protects a majority-women workforce (≈ 4 million workers, BGMEA); (3) **burden → career** — ad-hoc clerical compliance work, often done by women staff, becomes a titled, certified, visible career path.



Photo: "Garment factory in Bangladesh — women working", CC BY 2.0, Wikimedia Commons.

Porichoy DPP in a Box

Dashboard Evidence Method & sources বাংলা

Evidence base

Every paper below was retrieved from the Crossref API and its DOI is resolved live on: 2025-09-13. Titles, journals, years and authors are copied verbatim from the API response (never typed by hand). The base grounds Porichoy's problem statement (EU regulation, traceability, Bangladesh RMG compliance) and its design choices (circularity, data provenance).

DPP in regulation - 5 traceability technology - 2 Bangladesh RMG compliance - 1 Bangladesh circularity - 1 circular economy - 2 sustainable supply chain - 1

due diligence & buyers - 1 circularity & recycling - 1 textile waste impact - 1 DPP & tracking - 1

#	PAPER	VENUE	YEAR	GROUP	CITED BY	DOI (VERIFIED)
1	The Circular Economy – A new sustainability paradigm? Martin Gassmann, Paulo Sargat, Nancy M.P. Bocken et al.	Journal of Cleaner Production	2017	circular economy	7,922	10.1016/j.jclepro.2014.12.048
2	Conceptualizing the circular economy: An analysis of 114 definitions Julien Kuczkowski, Denise Reike, Marco Helbert et al.	Resources, Conservation and Recycling	2017	circular economy	7,289	10.1016/j.resconrec.2017.09.005
3	From a literature review to a conceptual framework for sustainable supply chain management Stefan Seuring, Martin Müller	Journal of Cleaner Production	2008	sustainable supply chain	4,930	10.1016/j.jclepro.2008.06.028
4	Blockchain-Enabled Supply Chain Traceability in the Textile and Apparel Supply Chain: A Case Study of the Fiber Producer, Lenzing Shihua Li, H. Armin Rezaei, Monica Carley	Sustainability	2021	traceability technology	94	10.3390/s131218495

The app's Evidence page: 16 verified references, publicly checkable.

7 · Pilot plan & risk register

PHASE (2026)	SCOPE	EXIT CRITERIA
Weeks 1–3 (Sep)	MVP live; deck, form answers, LOIs; submit NIC 3.0 by 30 Sep	Submission complete; 2–3 factory LOIs
Q4	Grant pilot: 10 factories, steward curriculum + certification, OCR for trim cards	10 factories onboarded; 20 stewards certified; mapping accuracy $\geq 90\%$ on real sheets
2027 H1	First paid subscriptions; one ERP API connector; brand-funded onboarding pilot	First revenue; readiness uplift documented per factory

RISK	MITIGATION
Track gate: open call targets "Existing Solution (1.5+ yrs traction)"	Honest staging via "Launched/Product-market fit"; strongest pilot evidence; work survives losing (real product + portfolio)
AI-content scrutiny of application texts	Applicant re-voices all prose before submission
Factory data access is slow	LOI template ready (EN+BN); one real messy sheet transforms the demo; bundled messy sample demonstrates the pipeline meanwhile
Delegated-act drift (rules change)	Versioned schema layer — regulatory updates are config

8 · References (all Crossref-verified 2026-09-13)

Retrieved from the Crossref REST API; metadata copied verbatim; each DOI resolved by direct request on 2026-09-13 and cached in the project citation record. Regulatory and market sources follow the reference list.

1. **[1]** Martin Geissdoerfer, Paulo Savaget, Nancy M.P. Bocken et al. The Circular Economy – A new sustainability paradigm?. *Journal of Cleaner Production*, 2017. DOI: [10.1016/j.jclepro.2016.12.048](https://doi.org/10.1016/j.jclepro.2016.12.048) · cited by undefined. Crossref-verified 2026-09-13
2. **[2]** Julian Kirchherr, Denise Reike, Marko Hekkert et al. Conceptualizing the circular economy: An analysis of 114 definitions. *Resources, Conservation and Recycling*, 2017. DOI: [10.1016/j.resconrec.2017.09.005](https://doi.org/10.1016/j.resconrec.2017.09.005) · cited by undefined. Crossref-verified 2026-09-13
3. **[3]** Stefan Seuring, Martin Müller From a literature review to a conceptual framework for sustainable supply chain management. *Journal of Cleaner Production*, 2008. DOI: [10.1016/j.jclepro.2008.04.020](https://doi.org/10.1016/j.jclepro.2008.04.020) · cited by undefined. Crossref-verified 2026-09-13
4. **[4]** Wafaa A. H. Ahmed, Bart L. MacCarthy Blockchain-Enabled Supply Chain Traceability in the Textile and Apparel Supply Chain: A Case Study of the Fiber Producer, Lenzing. *Sustainability*, 2021. DOI: [10.3390/su131910496](https://doi.org/10.3390/su131910496) · cited by undefined. Crossref-verified 2026-09-13
5. **[5]** Faheem Uddin Enhancing the Circular Economy in Textile Fiber Waste Recycling. *Materials Circular Economy*, 2025. DOI: [10.1007/s42824-025-00190-9](https://doi.org/10.1007/s42824-025-00190-9) · cited by undefined. Crossref-verified 2026-09-13
6. **[6]** Anne-Christin Mittwoch The Digital Product Passport of the Ecodesign Regulation - Passport to a Successful Twin Transformation in Product Law?. *Business Law Review*, 2024. DOI: [10.54648/bula2024009](https://doi.org/10.54648/bula2024009) · cited by undefined. Crossref-verified 2026-09-13
7. **[7]** Jeyran Bayramova, Steven Pires, Patricia A. Holden et al. Environmental impact of open burning of polyester and cotton textile waste: a comparative analysis. *Environmental Science and Pollution Research*, 2026. DOI: [10.1007/s11356-026-37563-9](https://doi.org/10.1007/s11356-026-37563-9) · cited by undefined. Crossref-verified 2026-09-13
8. **[8]** Katrin Dietrich, Tassilo Schuster, Lydia Bühler et al. Digital Product Passport: A Digital Catalyst for a Circular Transformation. *Journal of Circular Economy*, 2026. DOI: [10.55845/joce-2025-32212](https://doi.org/10.55845/joce-2025-32212) · cited by undefined. Crossref-verified 2026-09-13
9. **[9]** Lone Riisgaard Reclaiming Waste: Informal Circularity in Bangladesh and the Struggle for Green Value in the Apparel Value Chain. *Circular Economy and Sustainability*, 2026. DOI: [10.1007/s43615-026-00844-z](https://doi.org/10.1007/s43615-026-00844-z) · cited by undefined. Crossref-verified 2026-09-13
10. **[10]** Kévin Boutillier Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requirements for Interoperability. *Circular Economy and Sustainability*, 2026. DOI: [10.1007/s43615-026-01000-3](https://doi.org/10.1007/s43615-026-01000-3) · cited by undefined. Crossref-verified 2026-09-13
11. **[11]** Karin Kandananond The Integration of RFID and Blockchain Technology for the Supply Chain Traceability of Durians. *Production Engineering Archives*, 2026. DOI: [10.30657/pea.2026.32.13](https://doi.org/10.30657/pea.2026.32.13) · cited by undefined. Crossref-verified 2026-09-13
12. **[12]** Md. Mahabub Alam, Md. Shawan Uddin, Md. Sohel Rana et al. The Impact of Social Compliance Practices on the Ready-Made Garment Industry in Bangladesh. *Journal of Business Studies*, 2023. DOI: [10.58753/jbspust.4.1.2023.21](https://doi.org/10.58753/jbspust.4.1.2023.21) · cited by undefined. Crossref-verified 2026-09-13
13. **[13]** Juliet Lawal A cut above: A guide to improving due diligence in garments and footwear supply chains. *OECD Observer*, 2019. DOI: [10.1787/df7dc9d4-en](https://doi.org/10.1787/df7dc9d4-en) · cited by undefined. Crossref-verified 2026-09-13
14. **[14]** Xuejie Zhao A Sustainable Product Tracking System under the Combination of Textile Label Regulations and Digital Product Passport. *Textile & Leather Review*, 2026. DOI: [10.31881/tlr.2026.1234](https://doi.org/10.31881/tlr.2026.1234) · cited by undefined. Crossref-verified 2026-09-13
15. **[15]** Tanita Behrendt, Elisabeth Eppinger, Erika Zernickel et al. Enabling Sustainability Through Ecodesign and Recyclability in the Textile Industry: Understanding Material Compositions of Denim. *Journal of Circular Economy*, 2026. DOI: [10.55845/joce-2026-41321](https://doi.org/10.55845/joce-2026-41321) · cited by undefined. Crossref-verified 2026-09-13

16. **[16]** Aitor Aragón, Vincenzo Daponte Digital Product Logbook, the missing piece in the European Digital Product Passport puzzle. *Informes de la Construcción*, 2026. DOI: [10.3989/ic.7377](https://doi.org/10.3989/ic.7377) · cited by undefined. Crossref-verified 2026-09-13
17. **[R1]** European Commission — Ecodesign for Sustainable Products Regulation (ESPR) page: in force 18 Jul 2024; first Working Plan Apr 2025; customs automatic DPP checks. Fetched 2026-09-12. commission.europa.eu
18. **[R2]** CIRPASS-2 — EU co-funded DPP preparatory project; pilot deployments in textiles, electronics, tires, construction. Verified on cirpass2.eu 2026-09-13. cirpass2.eu
19. **[R3]** TrusTrace — brand-side DPP/traceability platform (category example). Fetched 2026-09-12. trustrace.com
20. **[R4]** BGMEA — industry statistics (≈4 million workers). bgmea.com.bd.

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